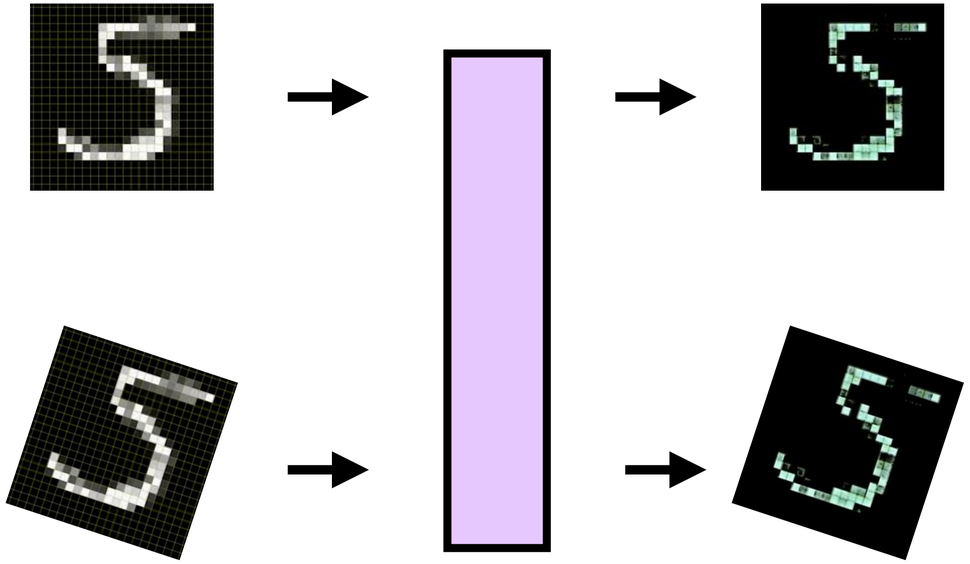
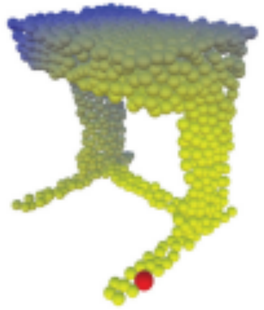
Geometric deep learning

- The key idea here is that constraining the network with translational symmetry dramatically narrows down the space of functions we optimize over, focusing on useful ones. **“Inductive bias”**.

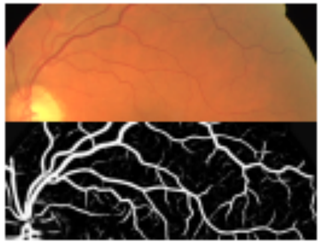
- CNN was for translations. It turns out we can do this for **any symmetry group *G*** (Cohen + Welling 2016): **group-equivariant neural networks.**



This framework — **geometric deep learning** — is a very useful thing to do! (Extensive formalism exists).



Wang et. al, 2019



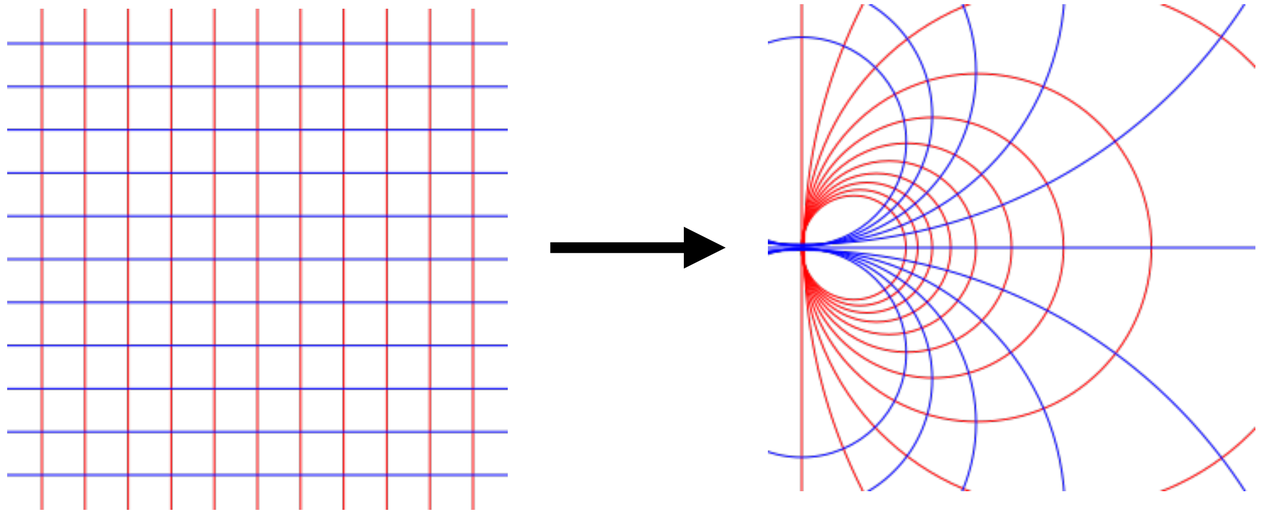
Bekkers et. al, 2018

Hoogeboom et. al, 2022

Conformal invariance

- I will now describe the first project of my **Pivot Fellowship**: how to make a neural network that is equivariant under **conformal invariance.**

- Recall conformal invariance is the set of coordinate transformations that **preserve angles**. In d dimensions, it is the group $SO(d+1,1)$.



$$x \to Rx + t$$ **Translations, rotations**

$$x \to \lambda x$$ **Scaling**

$$\frac{x}{|x|^2} \to \frac{x}{|x|^2} + b$$ **Special conformal**